

Blind Spot: An Interactive Gaze-aware Experience

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Figure 1: Screenshot: opening scene of the art installation *Blind Spot*.

ABSTRACT

Gaze in the context of interactive systems plays a vital role in Human-Computer Interaction (HCI). Gaze-aware systems provide support for users with physical disabilities, are employed as a game input device, or serve as an input form in public settings. The installation introduced in this work, called “Blind Spot”, reflects these developments by using gaze in an art context by partially obscuring the user’s field of view (i.e., the current gaze position). In doing so, the installation illustrates the potentials and characteristics of eye-tracking technology. It aims at fostering discussions about new design possibilities for gaze-based interactions in the arts and academia. Furthermore, it makes spectators aware of how

visual information is processed by humans (i.e., only a small fraction with high visual acuity is perceived at a given moment). Thirdly, it addresses the human’s desire to retrieve a complete picture of a given scenery (i.e., revealing this desire by limiting the spectator’s perception capabilities).

CCS CONCEPTS

• **Human-centered computing** → **Interaction techniques**; *Interaction devices*; HCI design and evaluation methods.

KEYWORDS

gaze interaction, art installation

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1 INTRODUCTION

A look says more than a thousand words: Gaze in the context of interactive systems plays a vital role in Human-Computer Interaction (HCI) [6, 8, 10, 13, 15, 18]. Gaze interaction supports users with physical disabilities (e.g., Galante and Menezes [2]) or may serve as an input technique in public settings (e.g., Khamis et al. [4]). The topic's relevance can also be seen in art projects. *EyeWriter* by Roth et al. [14], for example, enables people with physical disabilities to express themselves through eye-tracking technology creatively. In many cases, the gaze is used as an explicit interaction form by replacing or augmenting other input devices, such as mice, keyboards, or gamepads. However, only a minority of research projects explore alternative ways of gaze integration. Novel approaches, especially in the games' context, introduce gaze as an implicit or overt form of user guidance [5] or as enrichment for social encounters in virtual interpersonal constellations [17]. In this regard, researchers use the unique characteristics of human vision by incorporating peripheral vision [11], blinking behavior [9], or eye movements during closed lids in the interaction process [12]. The art installation *Blind Spot* critically discusses gaze-based interactions by experimenting with them via partially suppressing visual information. This work aims to foster conversations about new design possibilities for gaze-based interactions in the arts and academia.

In detail, *Blind Spot* addresses the following topics:

- it illustrates the potentials and characteristics of eye-tracking technology and aims at fostering discussions about new design possibilities for gaze-based interactions in the arts and academia
- it makes spectators aware of how visual information is processed by humans (i.e., only a small fraction with high visual acuity is perceived at a given moment)
- it addresses the human's desire to retrieve a complete picture of a given scenery (i.e., revealing this desire by limiting the spectator's perception capabilities)

2 ART INSTALLATION

In the art installation *Blind Spot*, the spectators' gaze is captured to hide elements located in the foveal vision region that is the only part of the retina that permits 100% visual acuity. In doing so, spectators solely obtain information via the peripheral vision system, which resides outside the point of fixation. In contrast to the foveal area, peripheral vision is weak in humans, especially at distinguishing detail, color, and shape. Its main functions are recognizing well-known structures and forms with no need to focus by the foveal line of sight, identifying similar forms and movements, and delivering sensations that form the background of accurate visual perception.

In the installation, the users find themselves in a video game-like scenario. The experience takes place in a wood that is designed to encourage exploration and serves as an indicator of the underlying themes through its setting (themes: being not able to see the wood for the trees, being lost in the woods, and seeking orientation). Spectators view the virtual world through their avatars' eyes (i.e., their representation within the virtual world) and control the

character by conventional means of input (gamepad). Via the eye-tracking device, however, they can only examine the objects that are located outside of the foveal region. This element is intended to arouse a feeling of unease, as curiosity and its (in our case: lack of) satisfaction are one of the critical attributes embedded in this exploration experience. This aspect is also transported through the installation's title and the question that is presented at the beginning: "What if we could not see the things that are in front of our very eyes?" We deem that the art installation may lead to further discussions within the HCI community on how gaze-based interactions could be designed. The installation reflects on the absence of visual information that could serve as a starting point for more in-depth exploration regarding the potentials and pitfalls of gaze-based interaction means.



Figure 2: Two in-game screenshots: when users look at an area on the screen, the section becomes white obstructing the visual information.

3 INTERACTION CONCEPT

Users interact with the virtual scenery via a gamepad and an eye-tracking device. Like in First-Person-Perspective games, the installation is perceived through the avatar's eyes. *Blind Spot* features standard controls, such as navigating through the woods via the gamepad's analog left thumbstick or change the current camera view by using the right thumbstick. The eye-tracking device continuously tracks the spectator's current view position. This information is used to occlude the current object of interest. The longer the spectator gazes at a specific area of a frame, the more the visual elements will be hidden. Thus, the spectator's character and gaze

have to be in constant movement to preserve the stimulus's visual integrity.

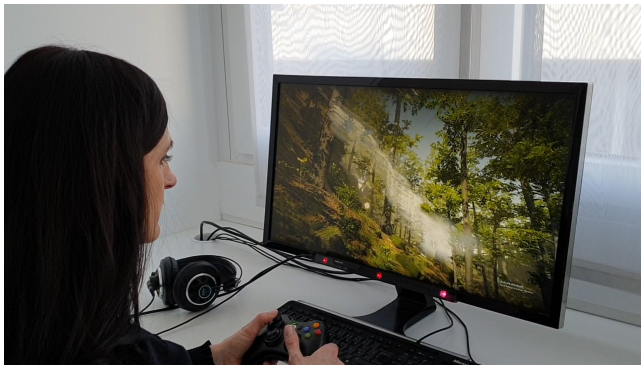


Figure 3: Technical setup: the installation consists of a a Tobii 4C eye tracker, a gamepad, a desktop PC, a monitor, and headphones.

4 TECHNICAL REALIZATION

Blind Spot is conceptualized as an art installation built in the Unity game engine [16]. It uses off-the-shelf eye-tracking technology (i.e., Tobii 4C [3]) that communicates with the game engine (i.e., Tobii EyeX SDK for Unity [1]). Image-based screen shaders were implemented to extract the currently gazed game object based on each frame's depth-buffer to achieve the occlusion-effect. Furthermore, users navigate through the game world via a Microsoft 360 gamepad. Concerning the experience's audiovisual presentation, the installation includes assets by NatureManufacture [7].

4.1 Technical Method

Two gaze visualization approaches were evaluated during the development process:

- **Global masking information** (UV-set with unique UV-layout for every triangle)
Advantages: possibility of durable obscurations even when returning to a region that was obscured
Disadvantages: much data to store, less flexibility - scene has to be built/pre-produced with a full unique UV-set
- **Screen-space masking information** (frame by frame)
Advantages: fast and flexible, only depends on frame related camera data
Disadvantages: as soon as obscured regions get out of sight (hidden by other objects or outside of camera frustum), the mask is erased, and so the region is visible again when coming into view the next time

Due to flexibility and ease of use, the screen-space-method was used for the project.

Short description of the used method (screen-space): A small circular cone from the viewer position opening in the viewing direction (detected by EyeX) is projected onto the scene. The surface that it first intersects is then colored white. So the information that part is obscured is stored in a mask where continuous obfuscation is

possible between 0.0 and 1.0. 0.0 sanding for clear sight, 1.0 for entirely hidden (here white). Together with the Depth-Buffer information, this gives a 3d-map of the obscured surfaces. The technical implementation is then independent of the content and relies only on the Depth-Buffer of the camera rendering, the projection frustum, and the camera position. The method works then on a frame by frame basis: For every frame, the previous frame's obscured region (or mask) is unprojected of the prior viewing parameters (camera frustum and position) and then reprojected onto the actual frame by using the actual camera frustum/position. The whole unprojection/projection math is implemented in a fragment shader that will be rendered as PostProcess-Effect on every frame.

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